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LASER PROCESSING APPARATUS AND LASER PROCESSING METHOD

Abstract

A laser processing apparatus includes: a holding unit configured to hold a workpiece; an oscillator configured to generate a laser beam; a splitter element configured to split the laser beam generated by the oscillator into a first laser beam and a second laser beam; a rotation mechanism configured to rotate the splitter element about an axis passing through a center of the splitter element; and a focusing unit including a focusing lens configured to focus the first laser beam split by the splitter element onto the workpiece held by the holding unit and to focus the second laser beam split by the splitter element onto the workpiece held by the holding unit.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S)

[0001] The present application claims priority to and incorporates by reference the entire contents of Japanese Patent Application No. 2024-030253 filed in Japan on Feb. 29, 2024.

BACKGROUND

[0002] The present disclosure relates to a laser processing apparatus and a laser processing method.

[0003] In order to improve productivity in laser processing, there has been proposed a laser processing apparatus capable of simultaneously processing a plurality of scheduled processing lines of a workpiece by simultaneously irradiating the plurality of scheduled processing lines with laser beams (refer to JP 2008-290086 A, for example).

[0004] The conventional processing apparatus has a configuration including a complicated mechanism for splitting a laser beam and a complicated mechanism for adjusting intervals between focusing points of split laser beams, having room for improvement.

SUMMARY

[0005] A laser processing apparatus according to one aspect of the present disclosure includes: a holding unit configured to hold a workpiece; an oscillator configured to generate a laser beam; a splitter element configured to split the laser beam generated by the oscillator into a first laser beam and a second laser beam; a rotation mechanism configured to rotate the splitter element about an axis passing through a center of the splitter element; and a focusing unit including a focusing lens configured to focus the first laser beam split by the splitter element onto the workpiece held by the holding unit and to focus the second laser beam split by the splitter element onto the workpiece held by the holding unit.

[0006] A laser processing method according to one aspect of the present disclosure is of processing a workpiece on which a plurality of scheduled processing lines extending in a first direction is set. The method includes: holding a workpiece by a holding unit; performing laser processing along the scheduled processing lines on the workpiece by splitting a laser beam generated by an oscillator into a first laser beam and a second laser beam by a splitter element, and relatively moving a focusing unit and the holding unit in a first direction while focusing light on the workpiece held by the holding unit using the first laser beam and the second laser beam, the performing the laser processing including positioning a focusing point of the first laser beam on a first scheduled processing line, positioning a focusing point of the second laser beam on a second scheduled processing line different from the first scheduled processing line, and emitting the first laser beam along the first scheduled processing line while emitting the second laser beam along the second scheduled processing line; and at least before performing the laser processing, adjusting an interval between the focusing point of the first laser beam and the focusing point of the second laser beam by rotating the splitter element based on a distance between the focusing point of the first laser beam and the focusing point of the second laser beam in a second direction orthogonal to the first direction.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] FIG. **1** is a perspective view illustrating a configuration example of a laser processing apparatus according to a first embodiment;

[0008] FIG. **2** is a perspective view illustrating a workpiece to be processed by the laser processing apparatus illustrated in FIG. **1**;

[0009] FIG. **3** is a diagram illustrating a configuration of a laser beam irradiation unit of the laser

processing apparatus illustrated in FIG. **2**;
[0010] FIG. **4** is a plan view illustrating an example of a focusing point of a first laser beam, a focusing point of a second laser beam, and the like in the laser beam irradiation unit illustrated in FIG. **3**;
[0011] FIG. **5** is a plan view illustrating another example of the focusing point of the first laser beam, the focusing point of the second laser beam, and the like in the laser beam irradiation unit illustrated in FIG. **3**;
[0012] FIG. **6** is a flowchart illustrating a flow of a laser processing method according to the first embodiment;
[0013] FIG. **7** is a plan view of the workpiece illustrating an example of a trajectory of the focusing point of the first laser beam and a trajectory of the focusing point of the second laser beam in a laser processing step of the laser processing method illustrated in FIG. **6**; and
[0014] FIG. **8** is a diagram illustrating a modification of the laser beam irradiation unit illustrated in FIG. **3**.

DETAILED DESCRIPTION

[0015] Preferred embodiments (embodiments) for carrying out the present disclosure will be described in detail with reference to the drawings. The present invention is not limited by the description in the following embodiments. In addition, the constituent elements described below include those that can be easily assumed by those skilled in the art and those that are substantially the same. Furthermore, the configurations described below can be appropriately combined with each other. In addition, various omissions, substitutions, or alterations in the configuration can be made without departing from the scope and spirits of the present invention.

First Embodiment

[0016] A laser processing apparatus according to a first embodiment of the present disclosure will be described with reference to the drawings. FIG. **1** is a perspective view illustrating a configuration example of a laser processing apparatus according to a first embodiment. FIG. **2** is a perspective view illustrating a workpiece to be processed by the laser processing apparatus illustrated in FIG. **1**. FIG. **3** is a diagram illustrating a configuration of a laser beam irradiation unit of the laser processing apparatus illustrated in FIG. **2**. FIG. **4** is a plan view illustrating an example of a focusing point of a first laser beam, a focusing point of a second laser beam, and the like of the laser beam irradiation unit illustrated in FIG. **3**. FIG. **5** is a plan view illustrating another example of the focusing point of the first laser beam, the focusing point of the second laser beam, and the like of the laser beam irradiation unit illustrated in FIG. **3**.

Workpiece

[0017] A laser processing apparatus **1** illustrated in FIG. **1** according to the first embodiment is a processing apparatus that processes a workpiece **200** illustrated in FIG. **2**. Examples of the workpiece **200** to be processed by the laser processing apparatus **1** illustrated in FIG. **1** according to the first embodiment include wafers such as a disk-shaped semiconductor wafer using a silicon substrate, a sapphire substrate, a gallium substrate, a SiC substrate, or the like as a substrate, or an optical device wafer.
[0018] In the first embodiment, as illustrated in FIG. **2**, the workpiece **200** has a configuration including a plurality of devices **203** formed in a region defined by scheduled processing lines **202-1**, being lines parallel to each other and parallel to a first direction **211** on a front surface **201**, and defined by scheduled processing lines **202-2**, being lines parallel to a second direction **212** orthogonal to the first direction **211**. In this manner, the plurality of scheduled processing lines **202-1** extending in the first direction **211** and the plurality of scheduled processing lines **202-2** extending in the second direction **212** are set in the workpiece **200**. The device **203** is, for example, an integrated circuit such as an Integrated Circuit (IC) or a Large Scale Integration (LSI), an image sensor such as a Charge Coupled Device (CCD) or a Complementary Metal Oxide Semiconductor (CMOS), or a memory device (semiconductor storage device).

[0019] In the first embodiment, the workpiece **200** is supported in an opening inside a frame **206** having an annular shape, by allowing a central portion of a tape **205** having a larger diameter than the workpiece **200** and having the frame **206** stuck to an outer edge, to stick to a back surface **204**, being a surface on a back side of the front surface **201**. In the present disclosure, the workpiece **200** is not limited to the form of being stuck to the tape **205**. In the present disclosure, the workpiece **200** need not have the scheduled processing line **202-1**, **202-2** or the device **203**, formed on the front surface **201**.

Laser Processing Apparatus

[0020] As illustrated in FIG. **1**, the laser processing apparatus **1** according to the first embodiment includes a holding unit **10**, a movement unit **30**, a laser beam irradiation unit **20**, an imaging unit (not illustrated), a cleaning unit **40**, a transfer unit **50**, and a controller **100**.

[0021] The holding unit **10** has a disk shape, and has a holding surface **11**, being flat in the horizontal direction and configured to hold the workpiece **200**, formed of a material such as porous ceramic. In addition, the holding unit **10** is movable by the movement unit **30** across two regions: a processing region, being a region below the laser beam irradiation unit **20**; and a loading/unloading region, being a region separated from the lower side of the laser beam irradiation unit **20** and used for loading/unloading of the workpiece **200**.

[0022] The holding unit **10** is connected to a vacuum suction source (not illustrated), and uses suction from the vacuum suction source to suck and hold the workpiece **200** placed on the holding surface **11**. In the first embodiment, the holding unit **10** sucks and holds the back surface **204** side of the workpiece **200** via the tape **205**. As illustrated in FIG. **1**, a plurality of clamping portions **12** for clamping the frame **206** is provided around the holding unit **10**.

[0023] The movement unit **30** relatively moves the holding unit **10** and the laser beam irradiation unit **20**. The movement unit **30** includes at least: a Y-axis movement unit **31**, being a stepped feed unit that moves the holding unit **10** in the Y-axis direction parallel to the horizontal direction; an X-axis movement unit **32**, being a processing feed unit that moves the holding unit **10** in the X-axis direction parallel to the horizontal direction and orthogonal to the Y-axis direction; and a rotary movement unit **33**, being a unit that rotates the holding unit **10** about an axis parallel to the Z-axis direction parallel to the vertical direction.

[0024] The Y-axis movement unit **31** is installed in an apparatus body **2**, and moves a movement plate **3** equipped with the X-axis movement unit **32** in the Y-axis direction, thereby moving the holding unit **10** in the Y-axis direction. The X-axis movement unit **32** is installed on the movement plate **3**, and moves a second movement plate **4** equipped with the rotary movement unit **33** in the X-axis direction, thereby moving the holding unit **10** in the X-axis direction. The rotary movement unit **33** is installed on the second movement plate **4** and supports the holding unit **10** to rotate the holding unit **10** about the axis.

[0025] The Y-axis movement unit **31** moves the X-axis movement unit **32**, the second movement plate **4**, the rotary movement unit **33**, and the holding unit **10** in the Y-axis direction, together with the movement plate **3**. The X-axis movement unit **32** moves the rotary movement unit **33** and the holding unit **10** in the X-axis direction, together with the second movement plate **4**.

[0026] The Y-axis movement unit **31** and the X-axis movement unit **32** include: a known ball screw rotatably provided around the axis; a known motor that rotates the ball screw around the axis; and a known guide rail that movably supports the movement plates **3** and **4** in the X-axis direction or the Y-axis direction. The rotary movement unit **33** includes a member such as a known motor that rotates the holding unit **10** about the axis.

[0027] As illustrated in FIG. **1**, the laser beam irradiation unit **20** has its partial portion provided at a distal end of a support column **6** whose proximal end portion is supported by an upright wall **5** erected from an end of the apparatus body **2** in the Y axis direction. The laser beam irradiation unit **20** irradiates the workpiece **200** held by the holding unit **10** with laser beams **28** and **29** (illustrated in FIG. **3**) to perform laser processing.

[0028] As illustrated in FIG. **3**, the laser beam irradiation unit **20** includes an oscillator **22**, a repetition frequency setting unit **23**, an output adjustment unit **24**, a splitter element **25**, a rotation mechanism **26**, and a focusing unit **27**. The oscillator **22** is a device that generates and oscillates a laser beam **21**, being a pulsed beam with a wavelength absorbable by the workpiece **200**. While the laser beam **21** has a wavelength absorbable by the workpiece **200** in the first embodiment, the laser beam may have a wavelength transparent to the workpiece **200** in the present disclosure, The repetition frequency setting unit **23** sets the repetition frequency of the laser beam **21** being a pulsed beam oscillated by the oscillator **22**.

[0029] The output adjustment unit **24** adjusts the output of the laser beam **21** oscillated by the oscillator **22**. In the first embodiment, the output adjustment unit **24** is a known attenuator. The splitter element **25** splits the laser beam **21** generated and oscillated by the oscillator **22** into a first laser beam **28** and a second laser beam **29**. In the first embodiment, the splitter element **25** splits the laser beam **21** such that the first laser beam **28** and the second laser beam **29** are aligned with each other in the Y-axis direction. In the first embodiment, the splitter element **25** includes a diffractive optical element (DOE) that splits one laser beam **21** into two laser beams, namely, the first laser beam **28** and the second laser beam **29**.

[0030] The rotation mechanism **26** rotates the splitter element **25** about an optical axis **251** of the laser beam **21** incident on the splitter element **25**, being an axis passing through the center of the splitter element **25**. In the first embodiment, the rotation mechanism **26** includes a motor or the like that rotates the splitter element **25** about the optical axis **251**.

[0031] The focusing unit **27** includes a focusing lens **271**, which focuses the first laser beam **28** split by the splitter element **25** on the workpiece **200** held by the holding unit **10** and focuses the second laser beam **29** split by the splitter element **25** on the workpiece **200** held by the holding unit **10**. In the first embodiment, the focusing lens **271** is a lens referred to as a biconvex lens. In the first embodiment, in addition to the focusing lens **271** being a biconvex lens, the focusing unit **27** further includes: a pair of known convex/concave lenses having convex surfaces facing each other so as to have the focusing lens **271** located between the convex/concave lens pair; a convex/concave lens located between the convex/concave lens on the holding unit **10** side and the focusing lens **271** and having a concave surface facing the focusing lens **271**; and a plano-convex lens located between this convex/concave lens and the convex/concave lens on the holding unit **10** side and having a convex surface on the focusing lens **271** side. The focusing unit **27** forms the first laser beam **28** to be perpendicular to the front surface **201** of the workpiece **200**, and forms the second laser beam **29** to be perpendicular to the front surface **201** of the workpiece **200**.

[0032] The laser beam irradiation unit **20** sets the focusing point **281** of the first laser beam **28** and the focusing point **291** of the second laser beam **29** formed by the focusing unit **27** to be beams aligned with each other in the Y-axis direction as illustrated in FIGS. **3** and **4**. Furthermore, the laser beam irradiation unit **20** has a configuration in which as the rotation mechanism **26** rotates the splitter element **25** about the optical axis **251**, thereby changing an interval **220** in the Y axis direction between the focusing point **281** of the first laser beam **28** and the focusing point **291** of the second laser beam **29**, formed by the focusing unit **27**, as illustrated in FIGS. **4** and **5**. That is, the interval **220** changes depending on an orientation **0** which is a rotation angle of the splitter element **25** around the optical axis **251**.

[0033] The imaging unit includes an imaging element that images a region to be split on the workpiece **200** before undergoing laser processing, held by the holding unit **10**. Examples of the imaging element include a charge-coupled device (CCD) imaging element or a complementary MOS (CMOS) imaging element. The imaging unit captures an image of the workpiece **200** held by the holding unit **10** so as to obtain an image for performing alignment, being an operation of aligning the workpiece **200** and the laser beam irradiation unit **20**, and outputs the obtained image to the controller **100**.

[0034] The cleaning unit **40** cleans the workpiece **200** after undergoing laser processing. The cleaning unit **40** has a disk shape, and includes a spinner table **41** and a cleaning nozzle **42**; the spinner table **41** is a table formed of a material such as porous ceramic and having a holding surface being flat in the horizontal direction and configured to hold the workpiece **200**. The spinner table **41** is rotated about an axis parallel to the Z-axis direction by a rotary drive source (not illustrated).

[0035] The spinner table **41** has its holding surface connected to a vacuum suction source (not illustrated), and sucks and holds the workpiece **200** placed on the holding surface on suction from the vacuum suction source. In the first embodiment, the spinner table **41** sucks and holds the back surface **204** side of the workpiece **200** via the tape **205**. In addition, a plurality of clamping portions **43** for clamping the frame **206** is provided around the spinner table **41**.

[0036] The cleaning nozzle **42** supplies cleaning water (in the first embodiment, pure water) to the front surface **201** of the workpiece **200** held by the spinner table **41** to clean the front surface **201** of the workpiece **200**.

[0037] The transfer unit **50** transfers the workpiece **200** between the holding unit **10** and the cleaning unit **40**. The transfer unit **50** includes a transfer arm **51** that transfers the workpiece **200** between the holding unit **10** and the cleaning unit **40**.

[0038] The controller **100** controls each component of the laser processing apparatus **1** to cause the laser processing apparatus **1** to perform processing operations onto the workpiece **200**. That is, the controller **100** controls at least the rotation mechanism **26**. The controller **100** is a computer including: an arithmetic processing apparatus having a microprocessor such as a central processing unit (CPU); a storage device having memory such as Read Only Memory (ROM) or Random Access Memory (RAM); and an input/output interface apparatus. The arithmetic processing apparatus of the controller **100** performs arithmetic processing according to a computer program stored in the storage device, and outputs a control signal for controlling the laser processing apparatus **1** to each component of the laser processing apparatus **1** via the input/output interface apparatus.

[0039] The controller **100** is connected to a display unit (not illustrated) including an apparatus such as a liquid crystal display apparatus that displays a state of a processing operation, an image, or the like, and to an input unit (not illustrated) used by an operator to register processing content information or the like. The input unit includes at least one of a touch panel provided on the display unit or an external input apparatus such as a keyboard.

[0040] As illustrated in FIG. **1**, the controller **100** includes a processing control unit **101**, an index value registration unit **102**, an information storage unit **103**, and an angle calculation unit **104**. The processing control unit **101** controls each component of the laser processing apparatus **1** to cause the laser processing apparatus **1** to perform processing operations onto the workpiece **200**.

[0041] The index value registration unit **102** registers an index value **213**. The index value **213** is an interval, in the Y-axis direction, between the position of the focusing point **281** of the first laser beam **28** focused on the front surface **201** of the workpiece **200** and the position of the focusing point **292** of the second laser beam **29** focused on the front surface **201** of the workpiece **200**. In the first embodiment, the index value **213** is an interval, in the Y-axis direction, between the focusing points **281** and **291** on the scheduled processing lines **202-1** adjacent to each other and the scheduled processing lines **202-2** adjacent to each other, on the front surfaces **201**.

[0042] The information storage unit **103** stores information **105**. The information **105** indicates a relationship between the orientation **0** (corresponding to the rotation angle) of the splitter element **25** around the optical axis **251** and the interval **220** between the focusing points **281** and **291** in the Y-axis direction. The angle calculation unit **104** calculates the orientation **0** of the splitter element **25** around the optical axis **251** based on the index value **213** registered in the index value registration unit **102** and the information **105** stored in the information storage unit **103**.

[0043] The functions of the index value registration unit **102** and the information storage unit **103** are implemented by the storage device. The functions of the processing control unit **101** and the angle calculation unit **104** are implemented by arithmetic processing performed by the arithmetic processing apparatus according to a computer program stored in the storage device.

Laser Processing Method

[0044] Next, a laser processing method according to the first embodiment will be described. FIG. **6** is a flowchart illustrating a flow of a laser processing method according to the first embodiment. FIG. **7** is a plan view of the workpiece illustrating an example of a trajectory of the focusing point of the first laser beam and a trajectory of the focusing point of the second laser beam in a laser processing step of the laser processing method illustrated in FIG. **6**.

[0045] The laser processing method is a laser processing method performed on the workpiece **200**, as a method of performing laser processing on the workpiece **200**. As illustrated in FIG. **6**, the laser processing method includes a holding step **1001**, a focusing point interval adjustment step **1002**, a laser processing step **1003**, and a cleaning step **1004**.

Holding Step

[0046] The holding step **1001** is a step of holding the workpiece **200** by the holding unit **10**. In the holding step **1001**, when the processing conditions are registered in the controller **100** by the operator or the like, the workpiece **200** is placed on the holding surface **11** of the holding unit **10** via the tape **205**, and the controller **100** receives an instruction to start the processing operation from the operator or the like, the laser processing apparatus **1** starts the processing operation, that is, the holding step **1001**. The processing condition includes the index value **213** and the like.

[0047] In the holding step **1001** performed by the laser processing apparatus **1** according to the first embodiment, the processing control unit **101** of the controller **100** sucks and holds the back surface **204** side of the workpiece **200** onto the holding surface **11** of the holding unit **10** via the tape **205**, and clamps the frame **206** using the clamping portion **12**.

Focusing Point Interval Adjustment Step

[0048] The focusing point interval adjustment step **1002**, being a step to be implemented at least before performing the laser processing step **1003**, is a step of adjusting the interval **220** between the focusing point **281** of the first laser beam **28** and the focusing point **291** of the second laser beam **29** by rotating the splitter element **25** based on the index value **213** being a distance between the focusing point **281** of the first laser beam **28** and the focusing point **291** of the second laser beam **29** in the second direction **212** orthogonal to the first direction **211**.

[0049] In the first embodiment, in the focusing point interval adjustment step **1002** performed by the laser processing apparatus **1**, the angle calculation unit **104** of the controller **100** calculates the orientation **0** of the splitter element **25** around the optical axis **251** based on the index value **213** registered in the index value registration unit **102** and the information **105** stored in the information storage unit **103**. In the first embodiment, in the focusing point interval adjustment step **1002** performed by the laser processing apparatus **1**, the processing control unit **101** of the controller **100** controls the rotation mechanism **26** to rotate the splitter element **25** around the optical axis **251** so as to achieve the orientation **0** around the optical axis **251** calculated by the angle calculation unit **104**, and sets the orientation **0** of the splitter element **25** to the orientation **0** calculated by the angle calculation unit **104**. In this manner, in the focusing point interval adjustment step **1002**, the processing control unit **101** of the controller **100** controls the rotation mechanism **26** based on the orientation **0** calculated by the angle calculation unit **104**.

Laser Processing Step

[0050] The laser processing step **1003** is a step including: splitting the laser beam **21** generated by the oscillator **22** into the first laser beam **28** and the second laser beam **29** by the splitter element **25**; and relatively moving the focusing unit **27** and the holding unit **10** in the first direction **211** while focusing light, by the focusing unit **27**, on the workpiece **200** held by the holding unit **10** using the first laser beam **28** and the second laser beam **29** so as to perform laser processing along the scheduled processing lines **202-1** and **202-2** on the workpiece **200**. In the first embodiment, in the laser processing step **1003** performed by the laser processing apparatus **1**, the processing control unit **101** of the controller **100** controls the movement unit **30** to move the holding unit **10** toward a processing region, captures an image of the workpiece **200** with the imaging unit, and performs alignment based on the image captured by the imaging unit.

[0051] In the first embodiment, in the alignment, the processing control unit **101** of the controller **100** adjusts the orientation of the holding unit **10** around its axis by the rotary movement unit **33** so that the first direction **211** of the workpiece **200** and the X-axis direction become parallel. Furthermore, in the first embodiment, in the alignment, the processing control unit **101** of the controller **100** controls the movement unit **30**, the focusing unit **27**, and the like to set the focusing point **281** of the first laser beam **28**, on which the laser beam **21** is split by the splitter element **25**, on the front surface **201** of any of the scheduled processing lines **202-1** among the plurality of scheduled processing lines **202-1** extending in the first direction **211**, and set the focusing point **291** of the second laser beam **29** on the front surface **201** of the scheduled processing line **202-1** adjacent to the scheduled processing line **202-1** on which the focusing point **281** of the first laser beam **28** has been set.

[0052] In the first embodiment, in the laser processing step **1003** of the laser processing apparatus **1**, the processing control unit **101** of the controller **100** controls the laser beam irradiation unit **20**, the movement unit **30**, and the like to relatively move the focusing unit **27** and the holding unit **10** of the laser beam irradiation unit **20** in the X-axis direction, that is, in the first direction **211**, so as to set the focusing points **281** and **291** of the laser beams **28** and **29** on the front surfaces **201** of the two scheduled processing lines **202-1** adjacent to each other on the workpiece **200**, and irradiates with the laser beams **28** and **29** the front surfaces **201** of the two scheduled processing lines **202-1** adjacent to each other and extending in the first direction **211** of the workpiece **200**. In this manner, in the first embodiment, in the laser processing step **1003**, the focusing point **281** of the first laser beam **28** and the focusing point **291** of the second laser beam **29** move on the front surfaces **201** of the two scheduled processing lines **202-1** adjacent to each other among the plurality of scheduled processing lines **202-1** extending in the first direction **211**, as illustrated in FIG. **7**.

[0053] In the first embodiment, in the laser processing step **1003**, when the laser processing apparatus **1** irradiates with the laser beams **28** and **29** the front surfaces **201** of the two scheduled processing lines **202-1** adjacent to each other, among the plurality of scheduled processing lines **202-1** extending in the first direction **211**, to apply ablation processing, and then, the processing control unit **101** of the controller **100** controls to relatively move the light focusing unit **27** of the laser beam irradiation unit **20** and the workpiece **200** held by the holding unit **10** in the Y-axis direction by a distance twice the index value **213** (hereinafter, referred to as indexed feed). Subsequently, the laser processing apparatus **1** irradiates with the laser beams **28** and **29** the front surfaces **201** of the next two scheduled processing lines **202-1** adjacent to each other and extending in the first direction **211** of the workpiece **200**.

[0054] In this manner, in the laser processing step **1003** in the first embodiment, the processing control unit **101** of the controller **100** repeats the irradiation of the laser beams **28** and **29** onto the front surfaces **201** of the two scheduled processing lines **202-1** adjacent to each other and extending in the first direction **211** and the indexed feed, so as to achieve irradiation with the laser beams **28** and **29** onto all the scheduled processing lines **202-1** extending in the first direction **211** of the workpiece **200**. Furthermore, in the laser processing step **1003** in the first embodiment, after irradiation with the laser beams **28** and **29** onto all the scheduled processing lines **202-1** extending in the first direction **211** of the workpiece **200**, the processing control unit **101** of the controller **100** controls the movement unit **30** to rotate the holding unit **10** by 90 degrees about the axis and irradiates with the laser beams **28** and **29** the scheduled processing lines **202-3** extending in the second direction **212**, similarly to the scheduled processing lines **202-1** extending in the first direction **211**. In the laser processing step **1003** in the first embodiment, the laser processing apparatus **1** irradiates with the laser beams **28** and **29** all the scheduled processing lines **202-1** and **202-2** of the workpiece **200** held by the holding unit **10** to form laser processed grooves (not

illustrated).
[0055] Hereinafter, of the scheduled processing line **202-1** and **202-2**, one of the scheduled processing line **202-1** and **202-2** simultaneously irradiated with the laser beams **28** and **29** in the laser processing step **1003** is referred to as a first scheduled processing line, and the other is referred to as a second scheduled processing line. For this reason, in the first embodiment, in the laser processing step **1003**, the focusing point **281** of the first laser beam **28** is positioned on the first scheduled processing line **202-1** and **202-2**, while the focusing point **291** of the second laser beam **29** is positioned on the second scheduled processing line **202-1** and **202-2** different from the first scheduled processing lines **202-1** and **202-2**, so that the first laser beam **28** is emitted along the first scheduled processing lines **202-1** and **202-2** while the second laser beam **29** is emitted along the second scheduled processing lines **202-1** and **202-2**.
Cleaning Step
[0056] The cleaning step **1004** is a step of cleaning the front surface **201** of the workpiece **200**. In the laser processing apparatus **1** in the cleaning step **1004**, the processing control unit **101** of the controller **100** controls units such as the movement unit **30** to move the holding unit **10** to the loading/unloading region, stops suction holding of the holding surface **11** of the holding unit **10** in the loading/unloading region, and stops clamping of the frame **206** of the clamping portion **12**.
[0057] In the laser processing apparatus **1** in the cleaning step **1004**, the processing control unit **101** of the controller **100** controls the transfer unit **50** to place the workpiece **200** from the holding unit **10** onto the holding surface of the spinner table **41** of the cleaning unit **40**. In the laser processing apparatus **1** according to the first embodiment in the cleaning step **1004**, the processing control unit **101** of the controller **100** sucks and holds the back surface **204** side of the workpiece **200** onto the holding surface of the spinner table **41** via the tape **205**, clamps the frame **206** using the clamping portions **43**, rotates the spinner table **41** about the axis, and drips the cleaning liquid from the cleaning nozzle **42** to the center of the workpiece **200** on the front surface **201** side.
[0058] The dripped cleaning liquid flows from the center side toward the outer peripheral side on the front surface **201** of the workpiece **200** by the centrifugal force generated by the rotation of the spinner table **41**, and cleans the front surface **201** of the workpiece **200**. In the cleaning step **1004**, the laser processing apparatus **1** according to the first embodiment supplies cleaning liquid for a predetermined time while rotating the spinner table **41** about the axis to clean the front surface **201** of the workpiece **200**.
[0059] As described above, the laser processing apparatus **1** and the laser processing method according to the first embodiment include: the splitter element **25** that splits the laser beam **21** generated by the oscillator **22**; and the rotation mechanism **26** that rotates the splitter element **25** about the optical axis **251** passing through the center of the splitter element **25**. With this configuration, the laser processing apparatus **1** and the laser processing method according to the first embodiment can change the interval **220** between the focusing points **281** and **291** of the laser beams **28** and **29** split by the rotation of the splitter element **25** about the optical axis **251** by the rotation mechanism **26**.
[0060] As a result, the laser processing apparatus **1** and the laser processing method according to the first embodiment have an effect of being able to adjust the interval **220** between the focusing points **281** and **291** of the laser beams **28** and **29** with a simple mechanism, namely, the rotation mechanism **26** that rotates the splitter element **25** about the optical axis **251**, and an effect of achieving simplification of the mechanism of adjusting the interval **220** between the focusing points **281** and **291** of the laser beams **28** and **29**.
[0061] In the present disclosure, the splitter element **25-1** is not limited to the diffractive optical element, and may be, for example, a polarizing prism that splits the laser beam **21** into the first laser beam **28** and the second laser beam **29**, such as a Rochon prism or a Wollaston prism as illustrated in FIG. **8**. FIG. **8** is a view illustrating a modification of the laser beam irradiation unit illustrated in FIG. **3**, and the same portions as those of the first embodiment are denoted by the

same reference numerals, and description thereof is omitted. The laser beam irradiation unit **20** illustrated in FIG. **8** is the same as that of the first embodiment except that the splitter element **25-1** is a polarizing prism.

[0062] In the present disclosure, the laser processing apparatus **1** may include a unit that forms a protective film on the front surface **201** of the workpiece **200** before irradiating the workpiece **200** with the laser beams **28** and **29**. In this case, it is desirable to coat the entire front surface **201** of the workpiece **200** with a protective film by applying a liquid water-soluble resin (for example, HogoMax (registered trademark) manufactured by DISCO Corporation) such as polyvinyl alcohol (PVA) or Polyvinylpyrrolidone (PVP) to the front surface **201** of the workpiece **200** and drying the liquid water-soluble resin. The unit for forming the protective film may be implemented by the cleaning unit **40** including a nozzle for supplying the liquid water-soluble resin to the front surface **201** of the workpiece **200**, or may be implemented by a unit different from the cleaning unit **40**.

[0063] According to the present disclosure, it is possible to achieve simplification of a mechanism of adjusting intervals between focusing points of laser beams.

[0064] Although the invention has been described with respect to specific embodiments for a complete and clear disclosure, the appended claims are not to be thus limited but are to be construed as embodying all modifications and alternative constructions that may occur to one skilled in the art that fairly fall within the basic teaching herein set forth.

Claims

1. A laser processing apparatus comprising: a holding unit configured to hold a workpiece; an oscillator configured to generate a laser beam; a splitter element configured to split the laser beam generated by the oscillator into a first laser beam and a second laser beam; a rotation mechanism configured to rotate the splitter element about an axis passing through a center of the splitter element; and a focusing unit including a focusing lens configured to focus the first laser beam split by the splitter element onto the workpiece held by the holding unit and to focus the second laser beam split by the splitter element onto the workpiece held by the holding unit.

2. The laser processing apparatus according to claim 1, wherein the splitter element includes a diffractive optical element.

3. The laser processing apparatus according to claim 1, wherein the splitter element includes a polarizing prism.

4. The laser processing apparatus according to claim 1, further comprising a controller configured to control at least the rotation mechanism, the controller including an index value registration unit configured to register an index value regarding an interval between a position of a focusing point of the first laser beam and a position of a focusing point of the second laser beam, and an angle calculation unit configured to calculate a rotation angle of the splitter element based on the index value registered in the index value registration unit, wherein the controller controls the rotation mechanism based on the rotation angle calculated by the angle calculation unit.

5. The laser processing apparatus according to claim 4, wherein a plurality of scheduled processing lines extending in a first direction is set on the workpiece, and the focusing point of the first laser beam is positioned on a first scheduled processing line, and the focusing point of the second laser beam is positioned on a second scheduled processing line different from the first scheduled processing line.

6. The laser processing apparatus according to claim 1, wherein the focusing unit forms the first laser beam in a direction perpendicular to the workpiece and forms the second laser beam in the direction perpendicular to the workpiece.

7. A laser processing method of processing a workpiece on which a plurality of scheduled processing lines extending in a first direction is set, the method comprising: holding a workpiece by a holding unit; performing laser processing along the scheduled processing lines on the workpiece by splitting a laser beam generated by an oscillator into a first laser beam and a second laser beam by a splitter element, and relatively moving a focusing unit and the holding unit in a first direction while focusing light on the workpiece held by the holding unit using the first laser beam and the second laser beam, the performing the laser processing including positioning a focusing point of the first laser beam on a first scheduled processing line, positioning a focusing point of the second laser beam on a second scheduled processing line different from the first scheduled processing line, and emitting the first laser beam along the first scheduled processing line while emitting the second laser beam along the second scheduled processing line; and at least before performing the laser processing, adjusting an interval between the focusing point of the first laser beam and the focusing point of the second laser beam by rotating the splitter element based on a distance between the focusing point of the first laser beam and the focusing point of the second laser beam in a second direction orthogonal to the first direction.